

CampusEats Dashboard Project Report

A detailed, screenshot-backed overview of the full multi-role business intelligence platform.

This project combines Streamlit dashboards, SQLAlchemy-backed data access, SQLite persistence, role-based authentication, and ML/AI features for campus food operations.

It includes a global command center, campus-level operations dashboard, stall-level analytics, a RandomForest-based sales forecaster, and a Gemini-powered business advisor.

The screenshots in this report were taken from the live app running inside the project venv, and they are saved in the Screenshots folder.

Core stack

- Streamlit UI and navigation
- SQLite database via SQLAlchemy
- Plotly and Folium visualizations
- Pandas / NumPy data processing
- scikit-learn forecasting
- Google Gemini chat assistant

Authentication & Data Layer

The app starts at Home.py with a two-step login flow. It loads environment variables, authenticates against the database, and keeps cached query helpers in database.py.

What this page shows

- Three-role login gateway: Global Admin, Campus Incharge, and Stall Owner.
- Demo mode accepts placeholder hashes, which is why the live screenshot can be reproduced without secrets.
- Session state stores user role, campus ID, entity ID, and display name for route control.
- SQLAlchemy provides the database engine; cached fetch helpers reduce repeated reads and keep reruns responsive.
- The project uses the attached SQLite file, so the dashboards render real figures instead of mock placeholders.

>



Business Intelligence Dashboard

Select your role to continue

 Global Admin

 Campus Incharge

 Stall Owner

Global Admin Dashboard

This page is the platform-wide command center. It summarizes overall revenue, active students, wallet top-ups, and the broader platform economy.

What this page shows

- Platform KPIs for revenue, active students, average order value, and wallet top-ups.
- 30-day revenue trend with a 7-day rolling average overlay.
- Hourly order and revenue pattern analysis.
- Order type, payment method, top stalls, and popular items breakdowns.
- Campus revenue map and table-style drilldowns for platform-wide management.

Home

Global Admin

Campus HQ

Stall Dashboard

AI Forecaster Advisor

CampusEats Dashboard

Global Command Center

Real-time snapshot of the entire CampusEats platform.

Total Platform ...

Active Students

Global Avg Ord...

Total Wallet To...

PKR 5...

999.0

PKR 5...

PKR 8...

Global admin overview, trend analysis, and platform benchmarking

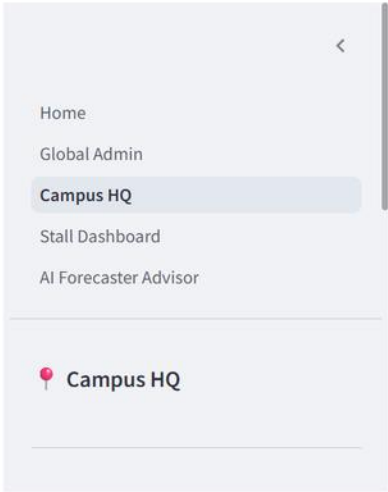
Generated from the live Streamlit app

Campus HQ Dashboard

Campus Incharge users get a campus-scoped operations view, with admin users able to switch campuses from the sidebar selector.

What this page shows

- Today's revenue, today's orders, active stalls, and available riders.
- Campus-specific 30-day revenue trend and category/payment breakdowns.
- Peak order hours chart for staffing and dispatch planning.
- Delivery partner performance and top student spender tables.
- Live campus map, stall markers, and rider markers with location context.
- A stall leaderboard plus an intervention center for cancellations and low ratings.



NUST — Campus HQ

Live operational overview for this campus.

 Today's Revenue	 Today's Orders	 Active Stalls	 Available Riders
PKR 0	0	33	30

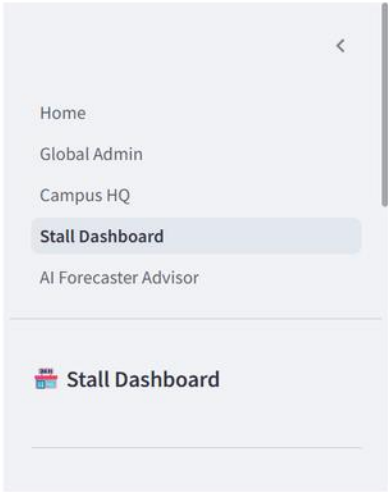
30-Day Revenue Trend

Stall Dashboard

The stall owner dashboard focuses on unit economics, menu performance, customer feedback, and item-level conversion problems.

What this page shows

- Financial analytics for today's subtotal, GST, tips, net today, all-time revenue, and completed orders.
- Rush-hour heatmap showing demand by hour and day of week.
- Menu intelligence with top-selling items and category revenue share.
- Dead weight report highlighting items frequently added to carts but rarely completed.
- Customer feedback section with rating distribution and review filtering.



Stall 100 (IBA Karachi) — Financial Analytics

Real-time earnings breakdown and 30-day revenue trends.

 Today's Subtotal  GST Collected  Tips Earned  Net Today

PKR 0 PKR 0... PKR 0... PKR 0

AI Forecaster

This module uses a RandomForestRegressor trained on historical orders to predict demand and recommend operational preparation for a specific day and hour.

What this page shows

- Prediction inputs include day of week, month, and a cumulative day counter.
- The page outputs predicted orders per hour, average revenue per order, and estimated hourly revenue.
- Top items likely to sell are inferred from historical hourly patterns.
- A 7-day forecast bar chart and historical daily-order chart show trend direction.
- Feature importances help explain what the model is using internally.

 **Super Admin**

Role:  Global Admin

 Sales Forecaster

☐  AI Business Advisor

Logout

Select Prediction Date

2026/05/04

Select Hour

12:00

Why did my revenue drop this week?

Which items should I promote next week?

What is my busiest day and how should I prepare?

How can I reduce my cancellation rate?

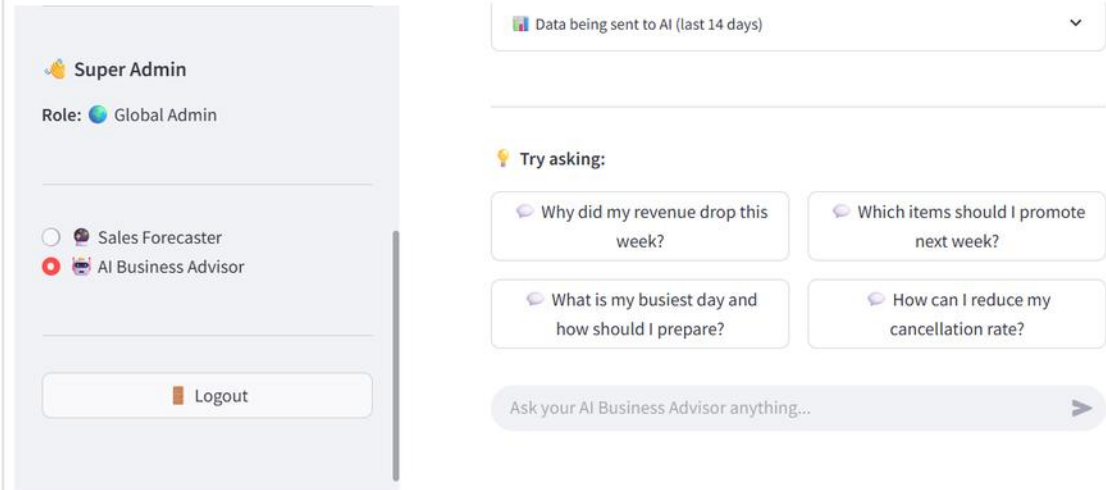
Ask your AI Business Advisor anything...

AI Business Advisor

The advisor page gathers 14 days of financial context, reviews, competitor benchmarks, and stall metadata before sending the prompt to Gemini.

What this page shows

- The advisor collects daily metrics, review summaries, and same-category competitors for context.
- Starter prompts help the user ask about revenue drops, busiest days, cancellations, and promotions.
- Response caching reduces repeated API calls within the same session.
- The code includes explicit handling for quota, API key, model-not-found, and safety-filter failures.
- This turns the dashboard into an explainable decision-support tool instead of a static reporting screen.



Project Notes & Validation

Validation summary

- The live app uses the project venv and the existing CampusEats.db database file.
- All screenshots were saved to the Screenshots folder and embedded into this report.
- The code base is organized around Home.py, database.py, and four role-driven pages under pages/.
- The dashboards emphasize real KPIs, campus operations, stall economics, forecasting, and AI-assisted analysis.
- The report was generated to be readable standalone, with enough detail to explain both what the app does and how the major views fit together.

Key files reviewed

- Home.py
- database.py
- pages/1_Global_Admin.py
- pages/2_Campus_HQ.py
- pages/3_Stall_Dashboard.py
- pages/4_AI_Forecaster_Advisor.py